Name _____ Period _____ Date _____

6-1 Lesson Quiz: Key Features of Exponential Functions

short form questions keep their Savvas numbers (2, 3, 4); show your work

2. 15,000 blue trout were released into the Meherrin River for a scientific study. The function $f(x) = 15{,}000\left(\frac{9}{8}\right)^x$ represents the number of blue trout after x years. After 5 years, what happens to the population?

Ⓐ It decreases by about 2,000.

Ⓑ It decreases by about 12,000.

Ⓒ It increases by about 2,000.

Ⓓ It increases by about 12,000.

Show your work: $f(5) =$ _____

Change after 5 years: _____

3. For the function $f(x) = \left(\frac{1}{2}\right) \cdot 6^x$, identify the domain, range, y-intercept, and asymptote. Write one of the given phrases or values in each blank.

linear	quadratic	exponential	all real numbers	0.5	0	0

Domain: _____ y-intercept: _____

Range: $\{y \mid y >$ _____$\}$ Asymptote: $y =$ _____

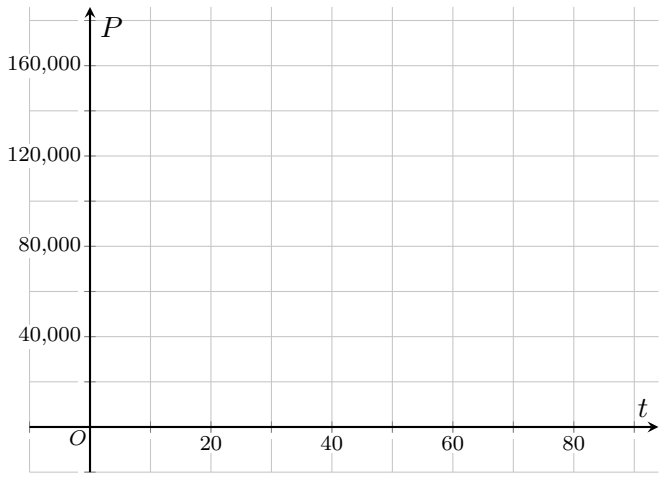
Function type: _____

4. The rabbit population of Springfield, Ohio, was 144,000 in 2016. It is expected to decrease by about 7.2% per year. Write and graph the exponential decay function of the rabbit population, $P(t)$.

$P(t) =$ _____ (_____)^t

Decay factor: _____

What does t count? _____



Scoring: Question 3 and Question 4 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

6-1 Lesson Quiz, short form: answer key

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2. D, it increases by about 12,000. The base $\frac{9}{8} = 1.125 > 1$, so the population grows. $f(5) = 15{,}000\left(\frac{9}{8}\right)^5 = 15{,}000 \cdot \frac{59{,}049}{32{,}768} \approx 27{,}030$, and $27{,}030 - 15{,}000 \approx 12{,}030$. Choice C is a common slip: $15{,}000(0.125) \approx 1{,}900$ is one year's increase, not five. Savvas quiz item 2, DOK 2, Skills Review and Practice F14. Taught by Example 3B (evaluate the model at a later time) and Try It 3b-c. **Bonus 5:** 3 for D, 2 for shown work ($f(5) \approx 27{,}000$ or the difference).

3. Domain: all real numbers; Range: $\{y \mid y > 0\}$**; y-intercept:** 0.5**; Asymptote:** $y = 0$**; Function type: exponential.** $a = \frac{1}{2}$ is the y-intercept because $6^0 = 1$; $b = 6 > 1$, so the graph rises from the x-axis on the left. Savvas quiz item 3, DOK 2, F13. Taught by Example 1 (Try It 1 is the same five features). **10 points:** 2 per correct blank.

4. $P(t) = 144{,}000(0.928)^t$**.** Decay factor $1 - 0.072 = 0.928$; t is years since 2016. Graph: starts at the filled point $(0, 144{,}000)$ and decreases toward the t-axis (asymptote $P = 0$), passing near $(10, 68{,}000)$, $(20, 32{,}000)$, $(40, 7{,}000)$; the TE answer graph ends near $(90, 0)$. Savvas quiz item 4, DOK 2, F13 and F14. Taught by Example 3 and the decay model box (Try It 3 is the same decay shape). **10 points:** 3 for 144,000, 4 for 0.928, 3 for a decreasing curve through $(0, 144{,}000)$ that approaches but does not cross the t-axis. A student who writes 0.072 as the base loses the 4 points (it is the rate, not the decay factor; concept summary Common Error).

Items left off this form (available for a retake or Form B): Savvas item 1, the function for $f(x) = 3^x$ after a vertical stretch by a factor of 8 and a reflection across the x-axis (answer D, $g(x) = -8 \cdot 3^x$; DOK 2, F28; it is the cut Example 2 and Lesson 1-2 transformations); Savvas item 5, which function has a greater growth factor than the graphed one through $(0, 16)$, $(2, 36)$, $(3, 54)$ (answer A, $f(x) = \left(\frac{3}{2}\right) \cdot 3^x$, because the graph's growth factor is $\frac{54}{36} = 1.5$ and $3 > 1.5$; choice B has the same factor 1.5; DOK 2, F14; it is the cut Example 5). Item 5's graph could be redrawn from the TE description; it is left off because no kept Example teaches it, not because of the figure (Example 1's study tip, ratio of consecutive y-values $= b$, is the one-minute bridge if the teacher puts it back).

IXL replacement: SmartScore 70 or above on EFD, U7P, DBW and TYQ replaces this quiz with 20/20.